



Society of Petroleum Engineers

SPE-227851-MS

Laser Deep Tunneling: An Alternative Waterless Technology to Hydraulic Fracturing

Abdullah Alharith, Sameeh Batarseh, Damian San Roman Algieri, and Ahmad Alrashed, Saudi Aramco PE&D

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227851-MS](https://doi.org/10.2118/227851-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This study provides experimental results of creating deep tunnels using high power laser technology as a waterless technology for hydraulic fracturing technology. The innovative technology has already been successfully deployed in the field to handle perforation jobs in hydrocarbon wells. The technology penetrated cased hole communicating the wellbore with the formation. The technology is a game changer that will provides safe and sustainable technology.

High power laser system with a 20 kW ytterbium-doped multiclad fiber laser with an emission wavelength of 1.07 microns is used for creating deep hole configurations in different types of rock for production applications in oil and gas wells. Recent developments in high power fiber lasers offer technical advantages when compared to other industrial lasers that may now allow economic subsurface applications to rock formations, delivering the beam through optical fiber from the surface via the wellbore. Successful applications in this manner would provide an alternative to conventional methods that employ hydraulic fracturing and shaped charge explosives.

Various parameters affecting laser penetration into samples of sandstone and limestone were studied and optimized. Deep tunnels are created reaching 64 inches in the lab, the tunnels can be extended deeper into the formation reaching over 100 feet in each direction. The tunnels can be created in all rock types with diameters up to 4 inches.

The success of the first deployment provided key elements and learning curve to create more advanced tunnels that can reach the targeted zone, the advantage of the laser is its stress and rock strength independent, it can create tunnels in any rock type regardless the strength. The technology can create communication with the hydrocarbon bearing formation allow the fluid to produce from the formation into the wellbore. This enable sustainable operation as the technology is waterless, cost effective and environmentally friendly technology. Laser can create enhanced tunnels with improved permeability.

The technology is waterless, cost effective and environmentally friendly, the innovative technology is stress and formation strength independent, eliminate the use of fresh water and reduce the surface footprint by reducing chemical, water tanks and pumps equipment.

Introduction

Stimulation techniques are common practices used to maintain hydrocarbon production in oil and gas reservoirs by improving permeability or creating new channels for hydrocarbons to flow through, reaching the wellbore. All types of reservoirs, including carbonate, sandstone, and shale, will require stimulation as the production rate declines.

The most common methods for enhancing productivity in carbonate reservoirs are matrix acidizing and hydraulic fracturing (Economides and Nolte, 2000). Hydraulic fracturing can be categorized into two main types: acid fracturing, and proppant hydraulic fracturing. In acid fracturing, the conductivity of the fracture is achieved through channels created by the dissolution of minerals by acids. In proppant hydraulic fracturing, the fracture is maintained by hard proppant materials, such as sand, or ceramic, which is injected into the fracture as part of a fracking fluids. For sandstone, or shale reservoirs, the rock matrix is relatively resistant to acid dissolution, making propped hydraulic fracturing the most effective method for achieving significant well stimulation in such reservoirs (Schwalbert et al., 2024).

Hydraulic fracturing involves pumping large amounts of fluids, including water, polymers, and chemical additives, along with proppant, into the wellbore under extremely high pressure (Saeed et al., 2023). These fluids are primarily composed of water as the base, polymers to thicken the fluid and carry the proppant, and various chemical additives that serve specific functions. While hydraulic fracturing provides a valuable source of energy, it raises environmental concerns, such as the impact on ecosystems and drinking water supplies. The use of chemicals in fracking fluids is particularly concerning, as they may pose risks to the environment and human health if they contaminate water sources. A significant amount of water is used in these operations, along with substantial quantities of proppant and chemicals, which pose considerable risks to Health, Safety, and the Environment (HSE).

Hydraulic Fracturing HSE Challenges

Water usage in fracking is substantial, ranging from 500,000 gallons to 6 million gallons per well, depending on the type of well and the number of stages treated. The quantity of proppant that are used for hydraulic fracturing, is directly related to water usage, as more proppant requires more water to transport it into the well (Alkhowaildi et al., 2021). These operations have potential health and environmental impacts, with water quality being one of the primary concerns. The fluids used in fracking, which are primarily water but also contain various chemicals and proppants, can pose risks if they contaminate drinking water supplies. These chemicals can be harmful to human health, potentially leading to illnesses if ingested.

Another critical issue is the management of wastewater generated from fracking operations. This wastewater is often contaminated with chemicals and other substances, raising concerns about safe disposal methods. Improper disposal could lead to soil and water contamination, further threatening environmental and public health (Adgate et al., 2014; Hill & Ma, 2021; National Institute of Environmental Health Sciences, 2025).

Hydraulic Fracturing Technical Challenges

In conventional hydraulic fracturing techniques, several parameters need to be considered, including permeability, rock hardness, mineralogy, young's modulus, and confining stress. Additionally, operational considerations such as injected fluids volume, proppant mass, injection rate, slurry proppant concentration, and gel viscosity must be taken into account (Schwalbert et al., 2024). Excessive water volume affects the production rate, causing formation damage. A high amount of injected fluids leads to high capillary pressure, affecting the relative permeability and blocking the fracture (Saeed et al., 2023). Providing the large amounts of water required for hydraulic fracturing, and then dealing with its disposal, is another challenge that comes at a significant cost (Scanlon et al., 2017).

High Power Laser for Upstream Oil and Gas Applications

Regardless of the HSE and technical challenges of hydraulic fracturing, high-power laser (HPL) technology offers a promising alternative to traditional hydraulic fracturing by enabling the creation of deep tunnels in downhole formations without the need for water, chemicals, or proppant. The advantages of this method include potential environmental benefits, as eliminating the use of water and chemicals will significantly reduce the environmental impact associated with hydraulic fracturing, disregarding the concerns related to water usage and chemical contamination. Additionally, HPL technology will eliminate the perforation operations that is required prior to hydraulic fracturing, as the laser can be used for both perforation and fracturing in a single operation. Furthermore, HPL technology requires a smaller footprint compared to what hydraulic fracturing needs. HPL technology presents an innovative approach to production stimulation, offering a cleaner and more efficient alternative to traditional methods. This could significantly reduce the environmental impact, lower operational costs, and enhance safety.

History of HPL Technology for Oil and Gas

Over the last few decades, scholars have increasingly explored the application of high-power lasers for subsurface applications in the oil and gas industry. Starting with fundamental studies of how the HPL will interact with different rock types, researchers have investigated the interaction between high-power lasers and various rock types, including carbonate, sandstone, and shale, to assess the effects of this technology on such geological formations (Batarseh et al., 2003; Gahan, 2001; Graves & O'Brien 1998; Graves, 2003; Reed, et al., 2003; Xu, et al., 2005). Such research studies open the window for further exploring of using HPL for oil and gas applications. Research in this field declined at that time due to the limited availability of HPL systems capable of achieving significant impacts for oil and gas applications. Years later, as the laser technology evolves, 10+ kilowatts (kW) HPL system become commercial and adorable (Zediker, 2014), leading to further research and development of HPL toward oil and gas subsurface applications. Such researches demonstrated the ability of HPL to be the future tool for perforation, and creating deep tunnel in the formation replacing the hydraulic fracturing rate (Batarseh et. al., 2019; Han et. al., 2019; Hu et. al., 2018).

The development of High-Power Laser (HPL) technology for subsurface applications in the oil and gas industry has shown promising results in laboratory testing, demonstrating its potential for improving efficiency and effectiveness in various rock formations. However, transitioning this technology from controlled lab settings to harsh field environment presents significant challenges, including system robustness, remote areas, power requirements, and downhole conditions. Recently, the scholars in this field made a groundbreaking achievement by successfully build the first HPL system capable of operating in downhole conditions, marking a new era in oil and gas industry (Batarseh et. al., 2023). This breakthrough not only addresses the technical challenges but also opens up possibilities for broader applications in subsurface oil and gas industry.

High Power Laser for Deep Tunneling

High power laser is a concentrated form of electromagnetic energy that converts into thermal energy when interacting with various types of materials. This thermal energy is harnessed to enable the laser to drill through different rock types, creating holes vary in size, shape, and depth depending on the amount of laser energy delivered to the rock. The process depends on HPL properties, including power, intensity, spot size, lasing time, and laser beam shape, as well as rock properties such as thermal conductivity, geological formation, and absorption rates.

Lab Testing and Evaluation

Several experimental works are conducted to study the ability of HPL to drill through various types of rocks. As demonstrated in Fig. 1, the power of this technology to achieve deep holes within a short time. A limestone sample with dimensions 20"x10"x10" (L×W×H) is used in the test, and a 10 kW laser is used to penetrate the limestone sample.

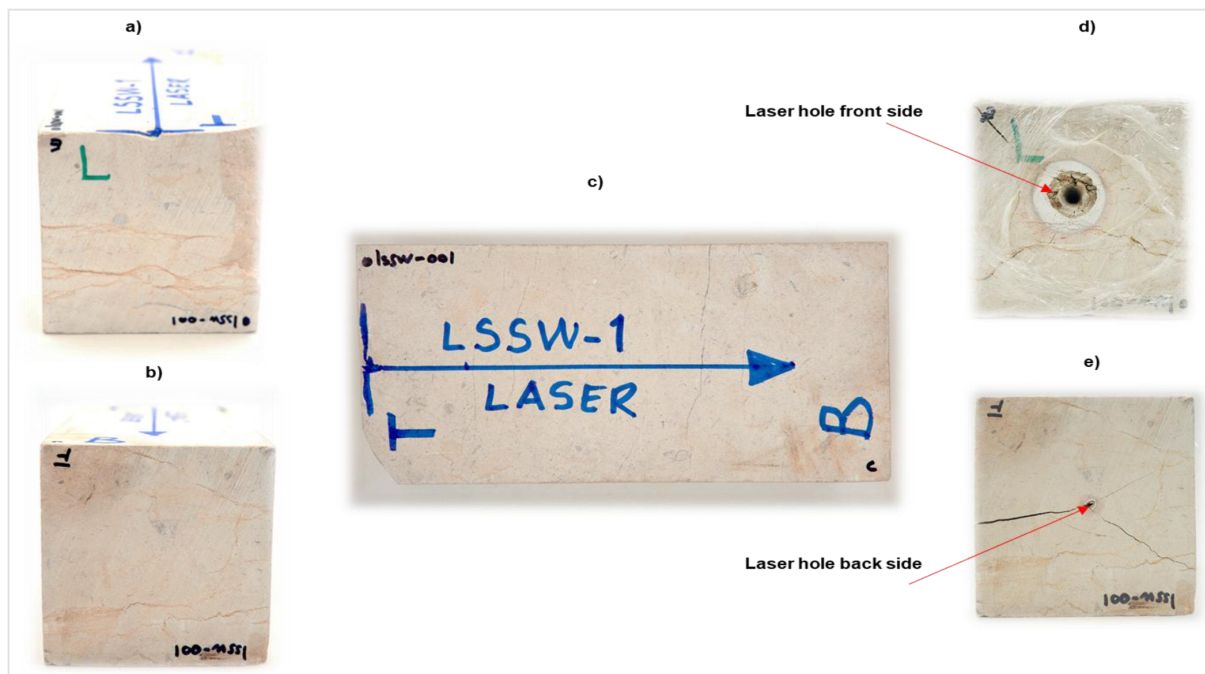


Figure 1—Limestone sample treated with HPL, a) front side before laser, b) Back side before laser, c) top view of 20 inches limestone sample, d) front side after laser, e) back side after laser.

Several physical and chemical changes are taking place when the HPL interacts with the rock sample, such changes include fragmentation, calcite decomposition, sublimation, and development of microcracks due to the thermal shock caused by HPL energy. The illustration in Fig.2. shows the shape of the tunnel created by the HPL and the microcracks that developed around it.

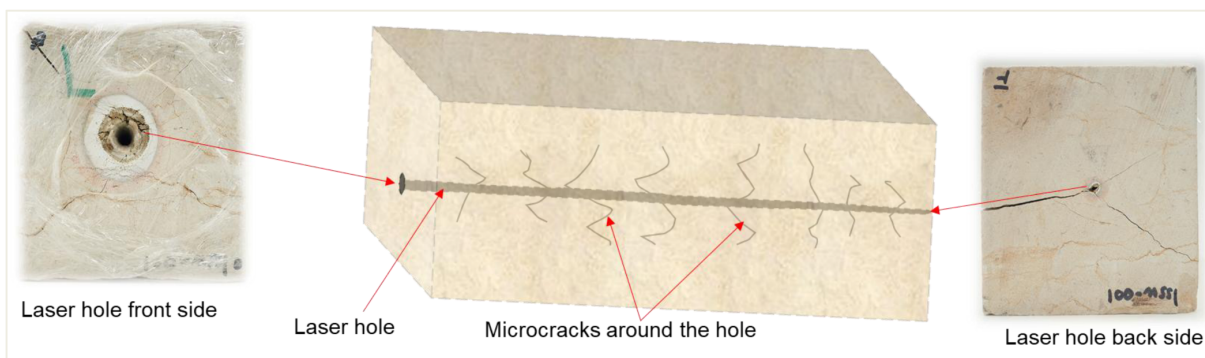


Figure 2—Tunnel made by HPL and microcracks around the tunnel.

Another test is carried out using a carbonate sample collected from a tight formation. The core sample measures 4 inches in diameter and 3 inches in length. This sample is treated with 10kW power, using a focused laser beam, and one inches spot size to create a hole through the sample. The laser achieves

penetration of the sample in few seconds (Fig. 3), highlighting the exceptional speed and efficiency of the HPL technology in such experiments.



Figure 3—Tight carbonate sample, a) top view of the sample, b) cross section of the sample.

Regardless of the rock type, HPL technology demonstrates a powerful means to drill through any type of formation overcome the limitations of conventional drilling methods, particularly in hard and abrasive rock types. Figure 4. illustrates the capability of HPL to create holes in any type of rock. The test was conducted on various rock types using a 10 kW HPL system. The samples included reservoir samples from both conventional and unconventional reservoirs, Berea sandstone samples, and a wide range of extremely hard igneous and sedimentary rocks (Batarseh et al.,2020).

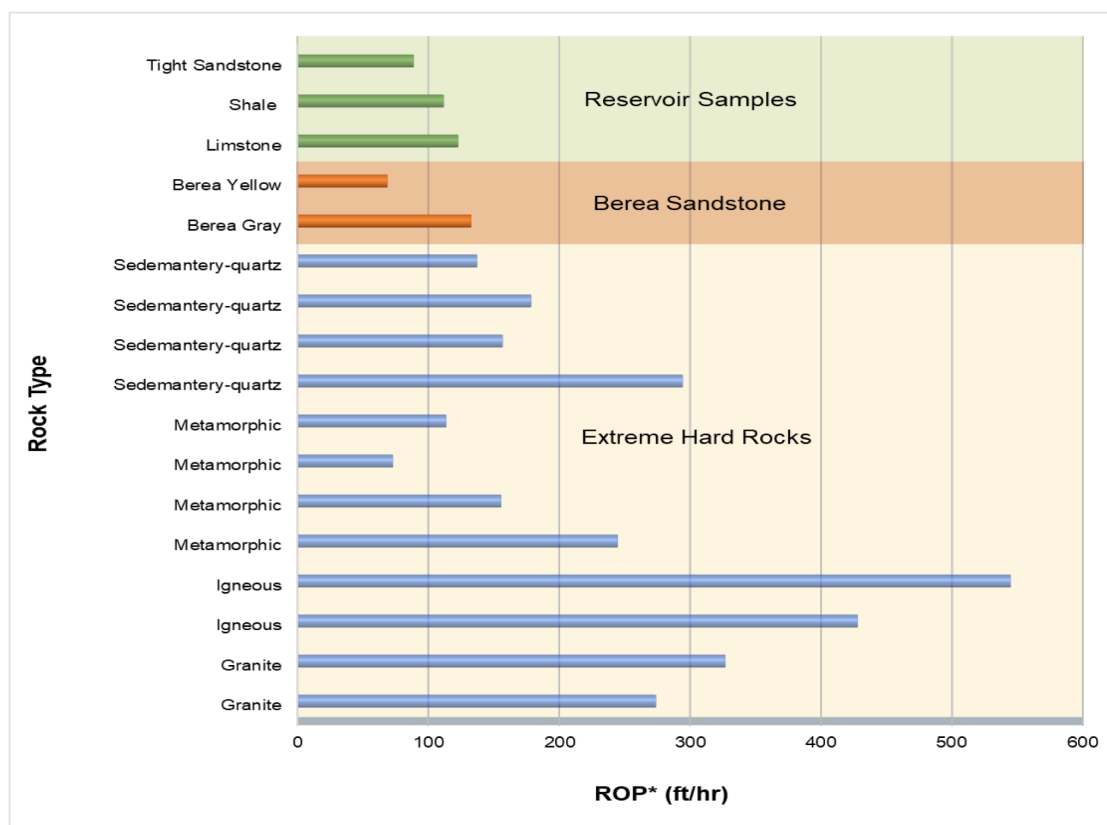


Figure 4—Extrapolate the rate of penetration (ROP) of HPL in various rock types. (ROP* extrapolate is only for illustration not to scale) (Batarseh et al.,2020).

High Power Laser for Field Application

The principle of HPL field operations is based on three main components: the surface equipment; the fiber optic cable, which is spooled into a coil tubing to reach downhole; and the bottom hole assembly (BHA), which is used to control and monitor the laser beam delivery to the formation.

HPL operations in the field require a small footprint and fewer number of equipment compared to the massive and complex equipment that are required for conventional hydraulic fracturing operations.

The surface equipment required to create a deep tunnel in the formation includes a laser generation unit, a coil tube unit, a nitrogen pumping tanker, and a control and operations room (Fig 5a). On the other hand, conventional hydraulic fracturing requires a large number of frack tanks to handle the large amounts of water, chemicals, and proppant, along with pumping equipment, complex piping system, and a large footprint (Fig 5b).

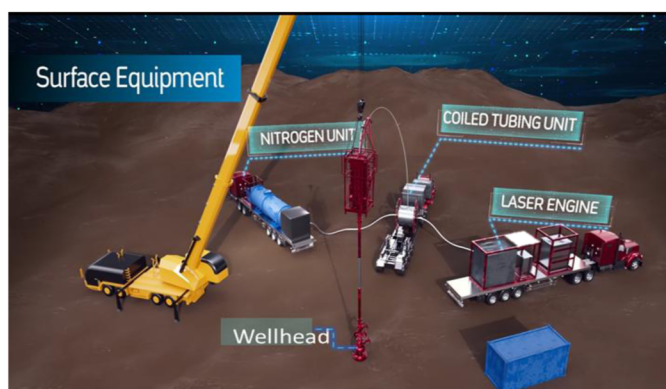


Figure 5a—Surface Equipment for HPL deep tunneling.



Figure 5b—illustration of surface equipment for hydraulic fracturing (Anisworth, 2017).

The wellsite preparations, manpower, and logistics for HPL operations are much less compared to what is required for hydraulic fracturing operations. HPL operations are more suitable for smaller or remote operations where mobilizing large amounts of equipment and resources is challenging. Thereby, the HPL is less resource-intensive, that offers potential cost savings.

Using High-Power Laser (HPL) technology to create deep tunnels in the formation will result in the development of several microcracks around the created tunnel. The thermal shock caused by the HPL energy induces residual stress around the tunnel, which generates microcracks and thereby improves the conductivity between the tunnel and the bearing formation nearby.

Remarks

As research in this field achieves a major milestone by enabling HPL to function in a downhole environment, a few further steps are required to create deep tunneling in the formation, which will replicate the objectives of a hydraulic fracturing operation.

The HPL technology offers several advantages, particularly in terms of environmental benefits and operational efficiency. One of the most notable benefits of this method is the elimination of the need for water, chemicals, and proppant, which are traditionally required in hydraulic fracturing operations. This reduction in resource usage not only lowers the environmental impact but also simplifies the logistics and costs associated with sourcing and transporting these materials.

Additionally, HPL technology eliminates the need for perforation operations, which are typically required prior to hydraulic fracturing. This further eco-friendly and sustainable process, reducing the time and resources needed to prepare the well for stimulation. The compact and efficient design of HPL systems represents a promising step forward in achieving more sustainable and efficient subsurface operations by minimizing the equipment and materials required, HPL technology has the potential to reduce the environmental footprint of oil and gas operation.

References

1. Adgate JL, Goldstein BD, McKenzie LM. Potential public health hazards, exposures and health effects from unconventional natural gas development. *Environ Sci Technol*. 2014;**48**(15):8307–8320. doi:[10.1021/es404621d](https://doi.org/10.1021/es404621d)
2. Ainsworth, K. "Automation of surface equipment, dissolvable and shorter plugs push efficiency of fracturing operations." *Drilling Contractor*, 2017, <https://drillingcontractor.org>. Accessed 29 May 2025.
3. Alkhawaili, Mustafa Ahmed, Mahmoud, Mohamed, Bataweel, Mohammed Abdullah, and Bassam Tawabini. "A Comprehensive Review on the Characteristics, Challenges and Reuse Opportunities Associated with Produced Water in Fracturing Operations." Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, November 2021. DOI: <https://doi.org/10.2118/207835-MS>
4. Batarseh, S. I., D. S. R. Alerigi, O. Al Obaid, and H. Othman. "Laser Gun: The Next Perforation Technology," 15 March 2019. <https://doi.org/10.2118/194775-MS>.
5. Batarseh, S., B. C. Gahan, R. A. Parker, R. M. Graves, and R. M. Graves, "Rock Phase Control by Using High Power Laser for Production Enhancements," no. 1, pp. 505, Jan. 2003. <https://doi.org/10.2351/1.5060060>.
6. Batarseh, Sameeh I., Alerigi, Damian P. San Roman, Marshall, Scott, and Ramona Graves. "First Industrial High-Power Laser Perforation Deployment." Paper presented at the SPE Annual Technical Conference and Exhibition, San Antonio, Texas, USA, October 2023. DOI: <https://doi.org/10.2118/215016-MS>
7. Batarseh, Sameeh I., Alharith, Abdullah M., Assiri, Wisam, and Damian San Roman Alerigi. "Laser Drilling in Extremely Hard Formation." Paper presented at the Offshore Technology Conference Asia, Kuala Lumpur, Malaysia, November 2020. DOI: <https://doi.org/10.4043/30416-MS>
8. Bridget R. Scanlon, Robert C. Reedy, Frank Male, and Mark Walsh. "Water Issues Related to Transitioning from Conventional to Unconventional Oil Production in the Permian Basin". *Environmental Science & Technology* 2017 **51** (18), 10903–10912. DOI:[10.1021/acs.est.7b02185](https://doi.org/10.1021/acs.est.7b02185)
9. Economides, M. J., and Nolte, K. G.: "Reservoir Stimulation", 3 Ed., London, UK, John Wiley & Sons, 2000.

10. Gahan, B.C., R.A. Parker, S. Batarseh, H. Figueroa, C.B. Reed, and Z. Xu. "Laser Drilling: Determination of Energy Required to Remove Rock." SPE-71466-MS. Presented at the SPE Annual Technical Conference and Exhibition, New Orleans, Louisiana, U.S.A., 30 September – 3 October 2001. <https://doi.org/10.2118/71466-ms>
11. Graves, R. M., "Application of High Power Laser Technology to Laser/Rock Destruction: Where Have We Been? Where Are We Now?" Jan. 2003.
12. Graves, R. M., and D. G. O'Brien. "StarWars Laser Technology Applied to Drilling and Completing Gas Wells." Paper presented at the SPE Annual Technical Conference and Exhibition, New Orleans, Louisiana, September 1998. DOI: <https://doi.org/10.2118/49259-MS>
13. Han, Y., Y. Fang, D. P. San-Roman-Alerigi, and S. I. Batarseh. "Thermal/Mechanical Interaction in Laser Perforation Process: Numerical-Model Buildup and Parametric Study." *SPE Journal* **24**, no. 5 (1 October 2019): 2097–2110. <https://doi.org/10.2118/183836-PA>.
14. Hill E, Ma L. The fracking concern with water quality. *Science*. 2021;**373**(6557):853–854. doi:[10.1126/science.abk3433](https://doi.org/10.1126/science.abk3433)
15. Hu, M., Y. Bai, H. Chen, B. Lu, and J. Bai. "Engineering Characteristics of Laser Perforation with a High Power Fiber Laser in Oil and Gas Wells." *Infrared Physics & Technology* **92** (1 August 2018): 103–8. <https://doi.org/10.1016/j.infrared.2018.05.014>.
16. National Institute of Environmental Health Sciences, "Hydraulic Fracturing & Health: Does hydraulic fracturing pose health risks to the people living near drilling sites?," *NIH*, <https://www.niehs.nih.gov/health/topics/agents/fracking>. Accessed May 13, 2025.
17. Reed, C. B., Z. Xu, R. A. Parker, B. C. Gahan, S. Bartarseh, R. M. Graves, H. Figueroa, and W. Deeg, "Laser Rock Drilling for Oil and Gas Wells: Abstract," Jan. 2003.
18. Saeed, Yousef Tareq, Alhajeri, Mubarak Muhammad, and Reza Barati Ghahfarokhi. "Investigation of Hydraulic Fracturing Sensitivity to Water Injection Volumes In Wolfcamp Formation." Paper presented at the Gas & Oil Technology Showcase and Conference, Dubai, UAE, March 2023. DOI: <https://doi.org/10.2118/214112-MS>
19. Schwalbert, M. P., Alves, I. N., and A. D. Hill. "Decision Criterion for Stimulation Method: Proppant Fracturing, Acid Fracturing, or Matrix Acidizing?." Paper presented at the SPE Annual Technical Conference and Exhibition, New Orleans, Louisiana, USA, September 2024. DOI: <https://doi.org/10.2118/220886-MS>
20. Xu, Z., Yamashita, Y., and C. B. Reed. "Modeling of Laser Spallation Drilling of Rocks for Gas– and Oil Well Drilling." Paper presented at the SPE Annual Technical Conference and Exhibition, Dallas, Texas, October 2005. DOI: <https://doi.org/10.2118/95746-MS>
21. Zediker, M. S. (11 April 2014). High-Power Fiber Lasers for Geothermal, *Oil, and Gas Industries*. SPIE Newsroom. <https://doi.org/10.1117/2.1201403.005288>